

Topical and Intralesional Cytotoxic Agents

5-FLUOROURACIL

Mechanism of Action and Formulations

Topical 5-fluorouracil (5-FU) has been used in clinical practice since the 1960s. It is a structural analogue of thymine that blocks DNA synthesis by inhibiting thymidylate synthetase. Topical 5-FU is commercially available as a 0.5 percent cream carried in a microsphere vehicle, as a 1 percent solution or 1 percent cream, as a 2 percent or 5 percent solution, and as a 5 percent cream. The solution can also be used for intralesional injection.

Indications

5-FU 5 percent cream has been approved by the U.S. Food and Drug Administration (FDA) since 1970 for the treatment of actinic keratoses in any location; 5-FU 0.5 percent cream is FDA approved for treatment of actinic keratoses of the face and anterior scalp. 5-FU is used twice daily until an inflammatory response is seen, usually 2 to 4 weeks. The 0.5 percent cream can be used once daily. 5-FU may also be a useful adjunct before cryosurgery of actinic keratoses. Pre-treatment with a keratolytic agent, prolonged therapy, more frequent applications, or the use of occlusive dressings may increase therapeutic response.

Treatment Considerations

Irritation during treatment sometimes requires reduction or interruption of treatment and the addition of emollients or topical corticosteroids to help quell brisk inflammation. A number of approaches to decrease irritation have been

devised; whether or not efficacy is preserved is disputed. Weekly pulse dosing involves applying the medication for 1 to 2 days per week for 6 to 7 weeks. Use of the 0.5 percent 5-FU cream with microsphere vehicle may allow for a more tolerable irritation profile.

During 5-FU treatment, previously sub-clinical actinic keratoses, basal cell carcinomas, and squamous cell carcinomas may respond to treatment and become apparent. In fact, the 5 percent 5-FU cream is approved by the FDA for the treatment of superficial basal cell carcinomas. Verrucae vulgaris, verrucae plana, plantar warts, and condylomata acuminata have been treated with varying response rates with topical and intralesional 5-FU. Additionally, actinic cheilitis, mucosal leukoplakia, radiodermatitis, Bowen disease, Bowenoid papulosis, and erythroplasia of Queyrat have all been reported to respond to treatment with topical 5-FU. Further, there are reports of the successful treatment of both psoriasis and keratoacanthomas with topical and intralesional 5-FU, and of infantile digital fibromatosis treated with intralesional 5-FU. Intralesional 5-FU has also been reported to be effective in the treatment of keloids and hypertrophic scars.

The systemic absorption of topically applied 5-FU is limited, with absorption selectively greater from abnormal skin than from surrounding normal skin. Microsphere-based formulations may retain 5-FU at the target site, with less drug reaching the systemic circulation. Rarely, systemic effects of 5-FU include nausea and anorexia, stomatitis, diarrhea, myelosuppression, alopecia, and cardiac and neurologic toxicity. Side effects of the topical preparation include local irritation, erythema, pain, swelling, pruritus, dyspigmentation, allergic contact dermatitis, and photosensitivity. Unusual reactions may include onycholysis, onychodystrophy, and the appearance of telangiectasias. The use of injectable preparations may be complicated by pain, local ulceration, erosion, desquamation, and dyspigmentation.

MECHLORETHAMINE (NITROGEN MUSTARD)

Mechanism of Action and Formulations

Mechlorethamine [methyl-bis (2-chloroethyl) amine hydrochloride] was the first nitrogen mustard (NM) to be used in clinical medicine.